

How Haptics Can Help Your Project

Most of us know haptic technology by its most common usage today—the feedback we get on a touchscreen smartphone when we interact with the on-screen elements. However, haptics also promises next-generation applications. Here is everything you need to know about haptics



The FreeForm 3D modeling system allows technicians to sculpt custom body parts by touch

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Fundamentally, haptics is any form of non-verbal communication involving touch. Shaking hands when you greet someone is a haptic custom originated from western countries. What we are going to discuss here is the technological aspect of using haptics. When you receive a call on your mobile phone while it is in silent mode, and feel that gentle buzzing in your pocket, that is haptic technology at work.

How does it work?

Haptic technology utilises haptic perception, which is the ability to

recognise objects by touch. While traditionally the object that was referred to was a physical object, modern-day implementations include the ability to recognise virtual objects such as those that exist in a world connected by the Internet of Things.

Haptics uses tactile feedback, executed by applying force or motion on the body of a person, in the form of vibrations or other similarly identifiable effects. The human somatosensory system is an impressively complex sensory system which comprises sensors that allow our body to feel touch, temperature, pain and movement. These sensors detect the force applied by the actuators in a haptics system and allow

the user to have an additional level of interaction with the device that he is working with.

Overall, it could be said that haptics allows a person to feel input from the device, other than the conventional methods of either seeing or hearing it. Thus, it adds further depth to the level of interaction between humans and machines.

Why integrate haptics in your project?

Haptic technology works by using different kinds of actuators to create vibrations and movements that are perceived as tactile feedback by the user. These haptic effects are enabled by the actuators that apply force on the skin so that the user can feel the force. A variety of actuators are available these days for use by a designer.

"Many technology companies such as Immersion, Texas Instruments (TI), Senseg and AAC Technologies have come up with a set of development tools to work with haptics—to design, develop and test haptics-based applications. As the technology is in a fairly nascent stage, not many players have resources available for learning in the field," explains Anand Tamboli, managing director, Knewron.

Eccentric rotating mass motors

The most commonly used (and the least expensive) form of actuators use electromagnetic technologies to form eccentric rotating mass (ERM) motors, which are off-centre weighted motors that apply very strong force. While the force that we can leverage from this device is impressive, you will not be able